HCIA-AI V4.0 Certification Practice Exam • Practice Set 2

Module 9: Practice Exam Set 2 (100 Advanced Scenario-Based Questions)

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Question 1

TRUE OR FALSE • 16 Points

In Decision Tree algorithms, post-pruning is executed before the decision tree begins growing, whereas pre-pruning is executed only after the tree is fully grown.

A. True
B. False

Question 2

TRUE OR FALSE • 16 Points

The K-Means clustering algorithm is highly sensitive to the initial selection of cluster centroids and can easily converge to local optima.

A. True
B. False

Question 3

TRUE OR FALSE • 16 Points

When a machine learning model performs exceptionally well on the training dataset but exhibits very poor accuracy on unseen test data, the model has high bias.

A. True
B. False

Question 4

TRUE OR FALSE • 16 Points

In Huawei MindSpore, a mindspore.Tensor object can be directly converted into a standard NumPy array using its .asnumpy() member method.

A. True
B. False

Question 5

TRUE OR FALSE • 16 Points

In Huawei Ascend AI processors, the Unified Buffer (UB) is used to temporarily store input and output tensor data for the Vector Unit and Cube Unit.

A. True
B. False

Page Answers: **Q1**: False • **Q2**: True • **Q3**: False • **Q4**: True • **Q5**: True

Question 6

TRUE OR FALSE • 16 Points

A single-layer perceptron can successfully classify linearly non-separable problems such as the XOR logic function.

A. True
B. False

Question 7

TRUE OR FALSE • 16 Points

In convolutional neural networks, Max Pooling layers contain learnable weight parameters that are updated via backpropagation during network training.

A. True
B. False

Question 8

TRUE OR FALSE • 16 Points

In Python, a dictionary key can be any mutable object, such as a Python list, dictionary, or set.

A. True
B. False

Question 9

TRUE OR FALSE • 16 Points

In a Generative Adversarial Network (GAN), the Generator aims to maximize the probability that the Discriminator makes a mistake, while the Discriminator aims to distinguish real data from fake data.

A. True
B. False

Question 10

TRUE OR FALSE • 16 Points

In Huawei ModelArts, ExeML is designed for experienced AI algorithm engineers who want to develop complex custom distributed training scripts from scratch.

A. True
B. False

Page Answers: **Q6**: False • **Q7**: False • **Q8**: False • **Q9**: True • **Q10**: False

Question 11

TRUE OR FALSE • 16 Points

Feature scaling (such as Min-Max normalization or Z-score standardization) does not alter the split decisions or prediction results of tree-based models such as Decision Trees.

A. True
B. False

Question 12

TRUE OR FALSE • 16 Points

Static word embeddings such as Word2Vec and GloVe generate dynamic, context-dependent word representations where polysemous words receive different vectors based on sentence context.

A. True
B. False

Question 13

TRUE OR FALSE • 16 Points

In deep neural networks, the standard ReLU activation function completely avoids the vanishing gradient problem across both positive and negative input domains.

A. True
B. False

Question 14

TRUE OR FALSE • 16 Points

The ATC (Ascend Tensor Compiler) tool provided by Huawei CANN can convert open-source framework models (such as ONNX, Caffe, and MindSpore) into offline models (.om) supported by Ascend processors.

A. True
B. False

Question 15

TRUE OR FALSE • 16 Points

In supervised machine learning, Logistic Regression is actually a classification algorithm used to predict continuous numeric target values.

A. True
B. False

Page Answers: **Q11**: True • **Q12**: False • **Q13**: False • **Q14**: True • **Q15**: False

Question 16

TRUE OR FALSE • 16 Points

In Huawei MindSpore, executing 'context.set_context(mode=context.GRAPH_MODE)' configures the framework to run in static graph mode for maximum operational efficiency and deployment performance.

A. True
B. False

Question 17

TRUE OR FALSE • 16 Points

The ROC curve plots the True Positive Rate (TPR) on the vertical axis against the False Positive Rate (FPR) on the horizontal axis across all possible classification decision thresholds.

A. True
B. False

Question 18

TRUE OR FALSE • 16 Points

Random Forest builds multiple decision trees using Bagging (bootstrap aggregating) and random feature subsampling, which primarily acts to reduce model variance.

A. True
B. False

Question 19

TRUE OR FALSE • 16 Points

In Huawei Ascend AI processor architecture, the Matrix Multiply Unit (Cube Core) is primarily responsible for scalar control flow execution and address calculations.

A. True
B. False

Question 20

TRUE OR FALSE • 16 Points

In Transformer architecture, Multi-Head Attention allows the network to jointly attend to information from different representation subspaces at different positions.

A. True
B. False

Page Answers: **Q16**: True • **Q17**: True • **Q18**: True • **Q19**: False • **Q20**: True

Question 21

TRUE OR FALSE • 16 Points

In deep learning multi-class classification problems, Categorical Cross-Entropy Loss is the most commonly used loss function.

A. True
B. False

Question 22

TRUE OR FALSE • 16 Points

For any real-valued input, the Sigmoid activation function maps it to an output within the interval (0, 1).

A. True
B. False

Question 23

TRUE OR FALSE • 16 Points

The classical ID3 Decision Tree algorithm uses the Gini Index as its fundamental splitting criterion.

A. True
B. False

Question 24

TRUE OR FALSE • 16 Points

The CART (Classification and Regression Trees) algorithm uses the Gini Index as its node impurity splitting criterion.

A. True
B. False

Question 25

TRUE OR FALSE • 16 Points

Dropout is a deep learning regularization technique that works by randomly setting a specified fraction of neuron activations to zero during training forward passes.

A. True
B. False

Page Answers: **Q21**: True • **Q22**: True • **Q23**: False • **Q24**: True • **Q25**: True

Question 26

SINGLE-ANSWER QUESTION • 16 Points

In Huawei's Ascend 910 AI processor, what is the maximum computing performance delivered for half-precision floating point (FP16)?

- **A.** 128 or 256 TFLOPS
- **B.** 512 or 1024 TFLOPS
- **C.** 64 or 128 TFLOPS
- **D.** 256 or 512 TFLOPS

Question 27

SINGLE-ANSWER QUESTION • 16 Points

In Convolutional Neural Networks, what is the primary role of 1x1 convolutions?

- **A.** Capturing long-range spatial context across large image dimensions
- **B.** Replacing spatial max-pooling subsampling operations completely
- **C.** Doubling height and width pixel resolution via bilinear upsampling
- **D.** Cross-channel interaction and feature dimensionality reduction

Question 28

SINGLE-ANSWER QUESTION • 16 Points

Which of the following ensemble learning algorithms is based on the Boosting methodology?

- **A.** Random Forest
- **B.** Extra Trees
- **C.** Bagging
- **D.** AdaBoost

Question 29

SINGLE-ANSWER QUESTION • 16 Points

In Huawei CANN software architecture, what is the core responsibility of TBE (Tensor Boost Engine)?

- **A.** Managing high-speed physical PCIe hardware bus communications
- **B.** Custom neural operator development, optimization, and code compilation
- **C.** Routing cloud REST web API tokens and tracking compute billing
- **D.** Ingesting raw camera video streams and decoding H.265 frames

Question 30

SINGLE-ANSWER QUESTION • 16 Points

In mathematical optimization, which optimizer maintains an exponential moving average of squared past gradients to adaptively scale the learning rate?

- **A.** Momentum
- **B.** RMSProp
- **C.** Batch GD
- **D.** Mini-batch SGD

Page Answers: **Q26**: D • **Q27**: D • **Q28**: D • **Q29**: B • **Q30**: B

Question 31

SINGLE-ANSWER QUESTION • 16 Points

Given an input image of size 32x32, a convolution kernel of size 5x5, stride of 1, and no padding (valid padding), what is the size of the output feature map?

- **A.** 30x30
- **B.** 27x27
- **C.** 28x28
- **D.** 32x32

Question 32

SINGLE-ANSWER QUESTION • 16 Points

Which of the following clustering algorithms does NOT require the practitioner to pre-specify the number of clusters (K)?

- **A.** Gaussian Mixture Models with Expectation-Maximization
- **B.** Density-Based Spatial Clustering of Applications
- **C.** Standard K-Medoids Partitioning Representative Algorithm
- **D.** Classic K-Means Centroid-Based Optimization Algorithm

Question 33

SINGLE-ANSWER QUESTION • 16 Points

In Huawei MindSpore, which dedicated package provides modules for dataset loading, data pipeline processing, and data augmentation?

- **A.** mindspore.ops
- **B.** mindspore.train
- **C.** mindspore.nn
- **D.** mindspore.dataset

Question 34

SINGLE-ANSWER QUESTION • 16 Points

What is the primary structural limitation of standard recurrent neural networks (RNNs)?

- **A.** Vanishing and exploding gradient issues over long sequential time horizons
- **B.** Inability to execute backpropagation through standard reverse autograd
- **C.** Extremely high memory fragmentation caused by static matrix dimensions
- **D.** Inability to process sequential inputs of variable token lengths

Question 35

SINGLE-ANSWER QUESTION • 16 Points

In an LSTM cell, which gating structure determines how much historical information from the previous cell state C_(t-1) should be discarded?

- **A.** Output Gate
- **B.** Input Gate
- **C.** Modulation Gate
- **D.** Forget Gate

Page Answers: **Q31**: C • **Q32**: B • **Q33**: D • **Q34**: A • **Q35**: D

Question 36

SINGLE-ANSWER QUESTION • 16 Points

Which structural mechanism in the Transformer network is designed to preserve the sequential order of words without using recurrent loops?

- **A.** Feed-Forward Network
- **B.** Multi-Head Attention
- **C.** Layer Normalization
- **D.** Positional Encoding

Question 37

SINGLE-ANSWER QUESTION • 16 Points

In Support Vector Machines (SVM), what are the critical training sample points that lie directly on the margin boundaries called?

- **A.** Support Vectors
- **B.** Cluster Centroids
- **C.** Outlier Vectors
- **D.** Boundary Vectors

Question 38

SINGLE-ANSWER QUESTION • 16 Points

In Python NumPy, what is the resulting array shape when an array of shape (4, 1) is added to an array of shape (1, 3) using broadcasting rules?

- **A.** (4, 3)
- **B.** (1, 3)
- **C.** (4, 1)
- **D.** (3, 4)

Question 39

SINGLE-ANSWER QUESTION • 16 Points

In Huawei ModelArts, what is the smart data labeling technology that uses pre-trained models to accelerate annotation?

- **A.** HiLens Edge Camera Deployment Pipeline
- **B.** ExeML Automated Training Script Parser
- **C.** Semi-Automated or Smart Annotation Engine
- **D.** Rule-based regular expression parser

Question 40

SINGLE-ANSWER QUESTION • 16 Points

In binary classification evaluation, if a model yields 90 True Positives, 10 False Negatives, 20 False Positives, and 80 True Negatives, what is the Recall?

- **A.** 0.75
- **B.** 0.90
- **C.** 0.80
- **D.** 0.82

Page Answers: **Q36**: D • **Q37**: A • **Q38**: A • **Q39**: C • **Q40**: B

Question 41

SINGLE-ANSWER QUESTION • 16 Points

Which of the following problems represents an Unsupervised Learning task?

- **A.** Segmenting store customers into behavioral clusters based on purchasing patterns without predefined labels
- **B.** Classifying incoming emails into spam or legitimate inbox messages using thousands of human-labeled records
- **C.** Estimating real estate sale valuations based on historical floor area and municipal sales registries
- **D.** Detecting fraudulent financial credit transactions based on verified historical fraudulent banking records

Question 42

SINGLE-ANSWER QUESTION • 16 Points

What is the training phenomenon called when gradient magnitudes grow exponentially during backpropagation, causing weights to oscillate or produce NaN values?

- **A.** Vanishing Gradient
- **B.** Dying ReLU
- **C.** Underfitting
- **D.** Exploding Gradient

Question 43

SINGLE-ANSWER QUESTION • 16 Points

Which landmark deep learning architecture first introduced Residual Skip Connections (x + F(x)) to enable training of networks exceeding 100 layers?

- **A.** VGGNet
- **B.** Inception-V3
- **C.** ResNet
- **D.** AlexNet

Question 44

SINGLE-ANSWER QUESTION • 16 Points

In Huawei MindSpore, what is the primary role of the 'mindspore.train.callback.ModelCheckpoint' callback?

- **A.** Visualizing real-time loss curves on external web dashboards
- **B.** Partitioning training tensors across multi-node GPU clusters
- **C.** Terminating model training if intermediate loss values diverge
- **D.** Periodically saving model weights and parameters to checkpoint files

Question 45

SINGLE-ANSWER QUESTION • 16 Points

In Huawei CANN software stack, which core engine is responsible for whole-graph compilation, memory planning, and operator stream scheduling on Ascend chips?

- **A.** ModelArts Studio
- **B.** Graph Engine
- **C.** MindSpore Lite
- **D.** TBE Compiler Engine

Page Answers: **Q41**: A • **Q42**: D • **Q43**: C • **Q44**: D • **Q45**: B

Question 46

SINGLE-ANSWER QUESTION • 16 Points

Which of the following mathematical activation functions produces outputs strictly in the range (-1, 1)?

- **A.** Softplus Activation Curve
- **B.** Hyperbolic Tangent
- **C.** Rectified Linear Unit
- **D.** Classic Sigmoid Function

Question 47

SINGLE-ANSWER QUESTION • 16 Points

In Python, what is the exact output of evaluating the expression '[1, 2, 3] * 2'?

- **A.** [1, 2, 3, 1, 2, 3]
- **B.** [2, 4, 6]
- **C.** [[1, 2, 3], [1, 2, 3]]
- **D.** TypeError: cannot multiply sequence by non-int

Question 48

SINGLE-ANSWER QUESTION • 16 Points

In Natural Language Processing, which pretrained language model uses a bidirectional Transformer encoder trained on Masked Language Modeling and Next Sentence Prediction?

- **A.** Word2Vec
- **B.** GPT-2
- **C.** BERT
- **D.** TextBlob

Question 49

SINGLE-ANSWER QUESTION • 16 Points

Which class of probabilistic machine learning models does the Naive Bayes algorithm belong to?

- **A.** Generative Model
- **B.** Reinforcement Learning Policy
- **C.** Discriminative Model
- **D.** Non-parametric Metric Model

Question 50

SINGLE-ANSWER QUESTION • 16 Points

In the Huawei Atlas AI product family, which hardware device is designed as a standard PCIe accelerator card plugged into servers for high-density inference?

- **A.** Atlas 200
- **B.** Atlas 900
- **C.** Atlas 300
- **D.** Atlas 500

Page Answers: **Q46**: B • **Q47**: A • **Q48**: C • **Q49**: A • **Q50**: C

Question 51

SINGLE-ANSWER QUESTION • 16 Points

What is the primary role of Principal Component Analysis (PCA) in machine learning?

- **A.** Reducing feature dimensionality while preserving as much data variance as possible
- **B.** Pruning deep decision branches to prevent overfitting in trees
- **C.** Normalizing raw model logits into a probability distribution summing to 1
- **D.** Measuring the harmonic mean between precision and recall

Question 52

SINGLE-ANSWER QUESTION • 16 Points

Which classification evaluation metric represents the harmonic mean of Precision and Recall?

- **A.** Area Under the ROC Curve
- **B.** Specificity
- **C.** F1-Score
- **D.** Accuracy

Question 53

SINGLE-ANSWER QUESTION • 16 Points

In Huawei MindSpore, which function is used to export a trained network model into the unified MindIR format for cross-platform deployment?

- **A.** mindspore.dump()
- **B.** mindspore.load()
- **C.** mindspore.save_checkpoint()
- **D.** mindspore.export()

Question 54

SINGLE-ANSWER QUESTION • 16 Points

In Computer Vision object detection, which of the following algorithms is a classic One-Stage (single-shot) detector?

- **A.** You Only Look Once Detector
- **B.** Region-Based Convolutional Net
- **C.** Fast R-CNN
- **D.** Faster R-CNN

Question 55

SINGLE-ANSWER QUESTION • 16 Points

What is the primary effect of adding L2 regularization (weight decay) into neural network training?

- **A.** Penalizing large weights by adding the sum of squared weights, encouraging small weights and lower variance
- **B.** Enforcing exact parameter sparsity by driving numerous individual weight coefficients precisely to zero
- **C.** Dynamically increasing the learning rate step size whenever validation accuracy plateaus during training
- **D.** Removing duplicated training data samples automatically from the training split during each epoch

Page Answers: **Q51**: A • **Q52**: C • **Q53**: D • **Q54**: A • **Q55**: A

Question 56

SINGLE-ANSWER QUESTION • 16 Points

Which of the following statements correctly highlights the core difference between Supervised and Unsupervised Learning?

A. Supervised learning optimizes continuous regression targets, whereas unsupervised predicts discrete classes

B. Unsupervised learning requires fully annotated target labels, whereas supervised operates only on unlabeled features

C. Supervised learning cannot be used for classification problems, whereas unsupervised learning handles multi-class targets

D. Supervised learning maps features to ground-truth labels, while unsupervised learning discovers intrinsic data structure

Question 57

SINGLE-ANSWER QUESTION • 16 Points

In Huawei Ascend AI processor DaVinci architecture, which specialized hardware unit executes dense 16x16x16 matrix multiplications in a single clock cycle?

A. Control Unit

B. Scalar Unit

C. Cube Unit

D. Vector Unit

Question 58

SINGLE-ANSWER QUESTION • 16 Points

In Reinforcement Learning, which function estimates the expected cumulative future reward starting from state s and following policy pi?

A. Value Function or Q-Function

B. Loss Function

C. Softmax Function

D. State Transition Matrix

Question 59

SINGLE-ANSWER QUESTION • 16 Points

In Python, which keyword is placed inside a function body to turn it into a generator function that yields values on demand?

A. pass

B. return

C. async

D. yield

Question 60

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following benefits are achieved by applying Batch Normalization in deep neural networks? (Multiple Choice)

A. It reduces internal covariate shift by normalizing layer activations across each mini-batch

B. It allows the use of greatly higher learning rates without causing training instability

C. It reduces the network's sensitive dependence on initial parameter weight values

D. It completely eliminates the need for activation functions in hidden layers

Page Answers: **Q56**: D • **Q57**: C • **Q58**: A • **Q59**: D • **Q60**: ABC

Question 61

MULTIPLE-ANSWER QUESTION • 18 Points

In Huawei ModelArts platform and MindSpore distributed architecture, which acceleration parallelism modes enhance large-scale model training efficiency?

- **A.** Data parallelism
- **B.** Pipeline parallelism
- **C.** Model pruning
- **D.** Tensor parallelism

Question 62

MULTIPLE-ANSWER QUESTION • 18 Points

In Linear Regression modeling, which of the following conditions or assumptions are evaluated?

- **A.** Linearity between independent explanatory features and the continuous target
- **B.** Homoscedasticity
- **C.** Absence of severe multicollinearity among explanatory input features
- **D.** Residuals are assumed to be approximately normally distributed around zero

Question 63

MULTIPLE-ANSWER QUESTION • 18 Points

In Machine Learning data preparation, which of the following approaches can be used to handle missing attribute values?

- **A.** Fabricating synthetic random continuous labels for unobserved training targets
- **B.** Predicting missing values using machine learning algorithms
- **C.** Deleting sample records or features that exhibit excessive missingness rates
- **D.** Imputing missing values with statistical measures

Question 64

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are hyperparameters configured prior to training rather than internal parameters learned via backpropagation?

- **A.** Learning Rate
- **B.** Number of Hidden Layers or Network Depth
- **C.** Weight Matrices of Convolutional Kernels
- **D.** Bias Vectors of Dense Output Neurons

Question 65

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following factors commonly contribute to Overfitting in deep neural network models?

- **A.** The training dataset is too small relative to model capacity
- **B.** Applying heavy L2 regularization and high dropout rates
- **C.** The model architecture has an excessively large number of parameters
- **D.** Training for too many epochs without early stopping mechanisms

Page Answers: **Q61**: ABD • **Q62**: ABCD • **Q63**: BCD • **Q64**: AB • **Q65**: ACD

Question 66

MULTIPLE-ANSWER QUESTION • 18 Points

In Huawei CANN software architecture, which of the following components are part of the compute software stack?

- **A.** Ascend Computing Language
- **B.** Fusion Engine
- **C.** Operator compilation tool
- **D.** Graph Engine

Question 67

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements regarding the K-Means clustering algorithm are TRUE?

- **A.** K-Means is sensitive to extreme outlier points
- **B.** The algorithm iterates between sample assignment to centroids and updating centroid coordinates
- **C.** K-Means results are completely fixed and independent of initial centroid placement
- **D.** K-Means assumes spherical clusters of approximately equal size and variance

Question 68

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following mechanisms are combined in the Adam optimizer to accelerate convergence?

- **A.** Momentum
- **B.** RMSProp
- **C.** Lasso
- **D.** Ridge

Question 69

MULTIPLE-ANSWER QUESTION • 18 Points

In Huawei MindSpore framework, which of the following capabilities are natively supported?

- **A.** Dynamic switching between PyNative and Graph execution modes
- **B.** Automatic Differentiation based on source-to-source transformation
- **C.** Device-Edge-Cloud synergy using unified MindIR representation
- **D.** Auto-Parallelism spanning data, model, and pipeline parallel modes

Question 70

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are foundational architectural building blocks of the Transformer model?

- **A.** Multi-Head Self-Attention mechanisms
- **B.** Position-wise Feed-Forward Neural Networks
- **C.** Layer Normalization and residual shortcut connections
- **D.** Bidirectional Recurrent Gated Memory Cells

Page Answers: **Q66**: ABCD • **Q67**: ABD • **Q68**: AB • **Q69**: ABCD • **Q70**: ABC

Question 71

MULTIPLE-ANSWER QUESTION • 18 Points

In Python, which of the following statements regarding functions are TRUE?

- **A.** Functions are first-class citizens and can be assigned to variables
- **B.** Functions can be passed as arguments into other functions
- **C.** Functions can be stored in container structures like lists and dictionaries
- **D.** Functions can return other functions as return values

Question 72

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following evaluation tools and metrics are specifically designed for continuous regression rather than classification?

- **A.** Mean Squared Error
- **B.** Coefficient of Determination
- **C.** Confusion Matrix Evaluation Grid
- **D.** Area Under the ROC Curve

Question 73

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements regarding Support Vector Machines (SVM) are correct?

- **A.** Standard kernel functions include Linear, Polynomial, and RBF kernels
- **B.** SVM finds an optimal hyperplane that maximizes the geometric margin between classes
- **C.** Kernel tricks map non-linearly separable inputs into higher-dimensional linear spaces
- **D.** All training data points contribute equally to determining the final decision boundary

Question 74

MULTIPLE-ANSWER QUESTION • 18 Points

In Huawei Atlas AI product family, which hardware devices are specifically engineered for cloud and data center large-scale training?

- **A.** Atlas 500
- **B.** Atlas 200
- **C.** Atlas 900
- **D.** Atlas 800

Question 75

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following characteristics differentiate Deep Learning from Traditional Machine Learning?

- **A.** Automatic end-to-end feature representation learning from raw data
- **B.** Superior performance scaling when trained on massive big datasets
- **C.** Supervised Gradient Backpropagation into Raw Features
- **D.** Guaranteed execution on low-power microcontrollers with zero compute

Page Answers: **Q71**: ABCD • **Q72**: AB • **Q73**: ABC • **Q74**: CD • **Q75**: AB

Question 76

MULTIPLE-ANSWER QUESTION • 18 Points

In Convolutional Neural Networks, which benefits are primarily achieved by applying Pooling (Subsampling) layers?

- **A.** Reducing spatial feature map dimensionality and parameter counts
- **B.** Providing spatial translation invariance to minor input shifts
- **C.** Expanding the effective receptive field of subsequent convolutional layers
- **D.** Converting floating-point tensors into ASCII text characters

Question 77

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following operations are standard data preprocessing steps in Natural Language Processing (NLP)?

- **A.** Lowercasing and punctuation text normalization
- **B.** Morphological stemming and lemmatization
- **C.** Autoregressive beam search decoding and temperature scaling
- **D.** Lexical tokenization and stopword removal

Question 78

MULTIPLE-ANSWER QUESTION • 18 Points

In Huawei ModelArts platform, which of the following lifecycle services are provided?

- **A.** Interactive development environment with cloud-hosted JupyterLab notebooks
- **B.** Multi-modal data ingestion, dataset management, and automated labeling
- **C.** Distributed model training with automatic hyperparameter optimization
- **D.** One-click deployment to real-time cloud REST APIs or edge devices

Question 79

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements regarding Gradient Descent algorithms are TRUE?

- **A.** Momentum helps accelerate convergence along shallow dimensions and suppresses oscillations
- **B.** Stochastic Gradient Descent updates parameters based on a single training example per step
- **C.** Mini-batch Gradient Descent balances vectorization speed and gradient estimation stability
- **D.** Standard SGD is mathematically guaranteed to converge monotonically without stochastic noise

Question 80

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following tasks are typical industrial applications of Computer Vision?

- **A.** Object detection and multi-object tracking
- **B.** Optical Character Recognition
- **C.** Syntactic dependency tree parsing in text datasets
- **D.** Semantic and instance image segmentation

Page Answers: **Q76**: ABC • **Q77**: ABD • **Q78**: ABCD • **Q79**: ABC • **Q80**: ABD

Question 81

MULTIPLE-ANSWER QUESTION • 18 Points

In Machine Learning, which of the following techniques can be used for dimensionality reduction?

- **A.** Linear Discriminant Analysis
- **B.** Principal Component Analysis
- **C.** Feature selection methods
- **D.** t-Distributed Stochastic Neighbor Embedding

Question 82

MULTIPLE-ANSWER QUESTION • 18 Points

In Huawei Ascend AI architecture, which of the following architectural properties describe the DaVinci NPU core?

- **A.** High-throughput 3D Cube computing engine optimized for matrix multiplication
- **B.** Unified Buffer architecture enabling high on-chip bandwidth data reuse
- **C.** Native hardware support for mixed-precision computation
- **D.** Dynamic clock throttling during all matrix multiplication operations

Question 83

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about the Sigmoid and Tanh activation functions are correct?

- **A.** The Sigmoid function output is zero-centered around 0
- **B.** The Sigmoid function compresses any input into the interval (0, 1)
- **C.** The Tanh function compresses input into the interval (-1, 1) and is zero-centered
- **D.** The ReLU function output is strictly confined to the range (-1, 1)

Question 84

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements regarding Decision Trees are TRUE?

- **A.** Decision trees strictly require all numerical inputs to be normalized to [0, 1]
- **B.** Small variations in training data can lead to completely different tree structures
- **C.** Deep unpruned decision trees have high variance and are prone to overfitting
- **D.** Decision trees are non-parametric and offer clear, interpretable decision logic

Question 85

MULTIPLE-ANSWER QUESTION • 18 Points

In Huawei MindSpore, which of the following statements regarding Tensors are correct?

- **A.** Tensors support slicing, tensor broadcasting, and mathematical operations
- **B.** Tensors in MindSpore cannot be converted back to NumPy ndarrays
- **C.** A Tensor is a multi-dimensional array data structure storing homogeneous data types
- **D.** Tensors can be instantiated directly from Python lists, tuples, or NumPy ndarrays

Page Answers: **Q81**: ABCD • **Q82**: ABC • **Q83**: BC • **Q84**: BCD • **Q85**: ACD

Question 86

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following architectural types represent foundational transformer attention variants in modern LLMs?

- **A.** CNN architecture
- **B.** RNN architecture
- **C.** Decoder-only architecture
- **D.** Encoder-decoder architecture

Question 87

MULTIPLE-ANSWER QUESTION • 18 Points

In LLM text generation, which decoding hyperparameters directly modulate output randomness and candidate token diversity?

- **A.** Temperature
- **B.** Top-P or Nucleus Sampling
- **C.** Training Epoch Iteration Count
- **D.** Learning Rate Warmup Decay Steps

Question 88

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following represent recognized main methods of Ensemble Learning in machine learning?

- **A.** Bagging
- **B.** Boosting
- **C.** Stacking or Blending
- **D.** Random Single-Tree Pruning without aggregation

Question 89

MULTIPLE-ANSWER QUESTION • 18 Points

In Python Object-Oriented Programming (OOP), which of the following statements are correct?

- **A.** The 'self' parameter explicitly references the specific class instance
- **B.** Python natively supports multiple inheritance
- **C.** The __init__ method acts as the class constructor executed during object instantiation
- **D.** Private attributes and methods are prefixed with double underscores

Question 90

MULTIPLE-ANSWER QUESTION • 18 Points

A Generative Adversarial Network (GAN) mainly consists of which competing neural subnetworks trained in a minimax game?

- **A.** Encoder
- **B.** Decoder
- **C.** Generator
- **D.** Discriminator

Page Answers: **Q86**: CD • **Q87**: AB • **Q88**: ABC • **Q89**: ABCD • **Q90**: CD

Question 91

MULTIPLE-ANSWER QUESTION • 18 Points

In Huawei Ascend computing, which of the following capabilities are provided by the AscendCL (Ascend Computing Language) API?

- **A.** Loads, initializes, and executes offline model files
- **B.** Provides unified C/C++ and Python programming interfaces
- **C.** Provides stream management APIs for asynchronous operator dispatch and synchronization
- **D.** Manages Device memory allocation, copy, and deallocation

Question 92

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following architectural strategies help prevent or reduce the Vanishing Gradient problem in deep networks?

- **A.** Using proper weight initialization techniques
- **B.** Using Residual Skip Connections
- **C.** Using non-saturating activation functions such as ReLU and Leaky ReLU
- **D.** Initializing all network weights to identical constant zero values

Question 93

FILL-IN-THE-BLANK QUESTION • 16 Points

On a convolution layer of a convolutional neural network, if there are 128 3 x 3 convolution kernels and the size of the input feature map is 28 x 28 x 64, the depth of a convolution kernel is ____. (Enter a number.)

Answer: _____

Question 94

FILL-IN-THE-BLANK QUESTION • 16 Points

CNN stands for ____. (Enter the full name.)

Answer: _____

Question 95

FILL-IN-THE-BLANK QUESTION • 16 Points

The ____ algorithm is based on the idea that if most of a sample's k nearest neighbors in the feature space belong to the same class, the sample also belongs to that class. (Enter an abbreviation.)

Answer: _____

Page Answers: **Q91**: ABCD • **Q92**: ABC • **Q93**: 64 • **Q94**: Convolutional Neural Network • **Q95**: KNN

Question 96

FILL-IN-THE-BLANK QUESTION • 16 Points

In multi-class classification tasks, the sum of all output probability values of the Softmax function is ____. (Enter a number.)

Answer: _____

Question 97

FILL-IN-THE-BLANK QUESTION • 16 Points

In k-fold cross-validation, the existing dataset is divided into ____ groups.

Answer: _____

Question 98

FILL-IN-THE-BLANK QUESTION • 16 Points

The three essential elements for AI development are data, compute, and ____.

Answer: _____

Question 99

FILL-IN-THE-BLANK QUESTION • 16 Points

A/An ____ is a binary classification model that is based on a linear classifier defined in the feature space that maximizes the margin. (Enter an abbreviation.)

Answer: _____

Question 100

FILL-IN-THE-BLANK QUESTION • 16 Points

In deep learning optimization, the technique that normalizes layer inputs across each mini-batch during training is called ____ Normalization.

Answer: _____

Page Answers: **Q96**: 1 • **Q97**: k • **Q98**: algorithms (or algorithm) • **Q99**: SVM • **Q100**: Batch